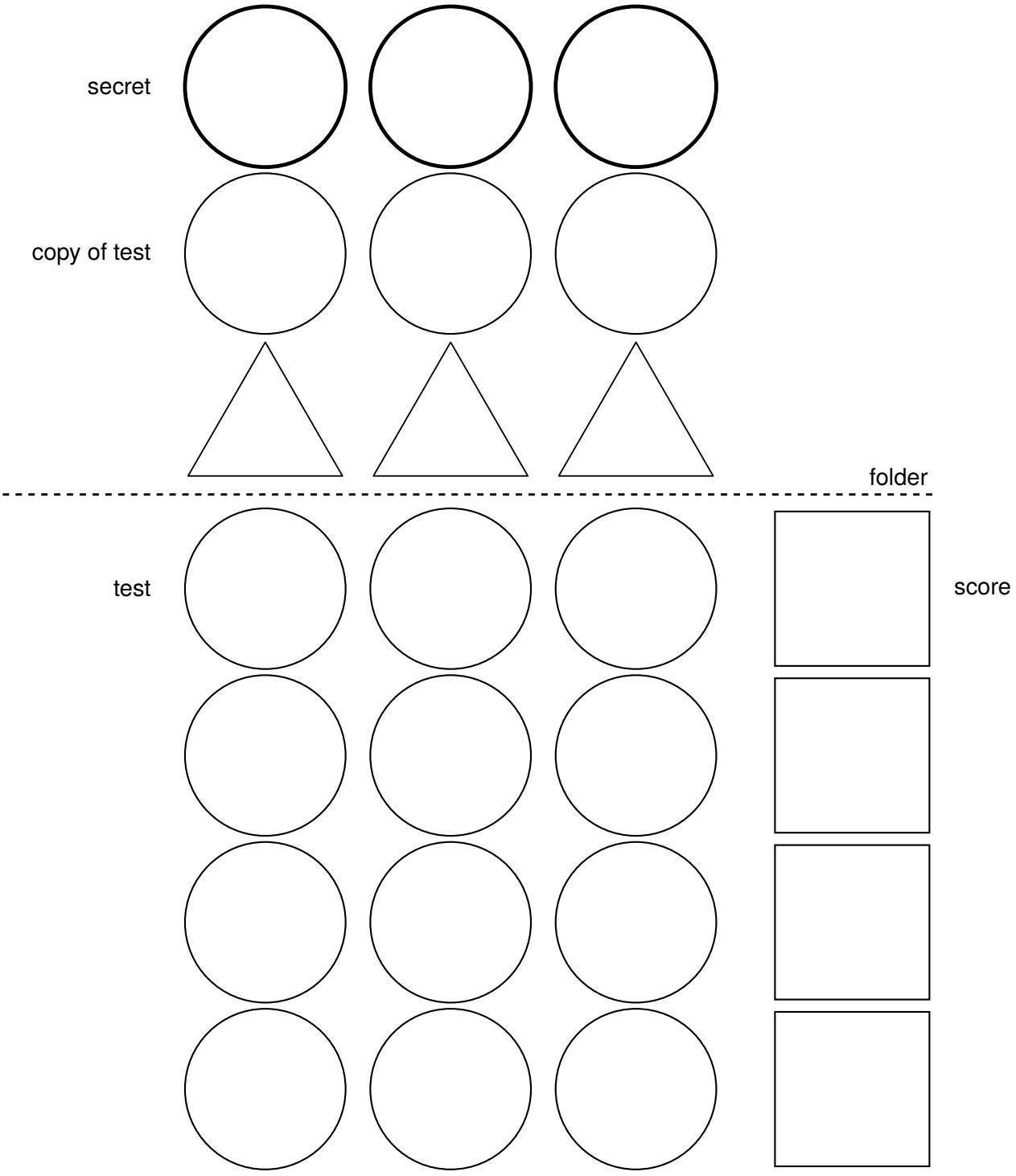
Problem 1: Your partner thinks of a whole number from 1 to 16. Ask yes-or-no questions until you are sure of the number. What is the smallest number of questions that always works? Explain why fewer questions cannot always work. Then answer the same questions for a number from 1 to 20.

The keeper hides a red/yellow secret row. The breaker fills a test row with red/yellow counters. Behind the folder, the keeper copies the test below the secret and puts a green triangle under each matching place. The breaker records the number of triangles as the score. After a final guess, uncover the secret and check the scores.

Problem 2: Play the code game with three counters, taking turns. Find each secret with as few tests as you can.



Problem 3: For each of these games, find every secret that fits all the scores. If no secret fits, explain how that could happen.

			score
R	R	Y	2
R	Y	R	2
Y	R	R	2

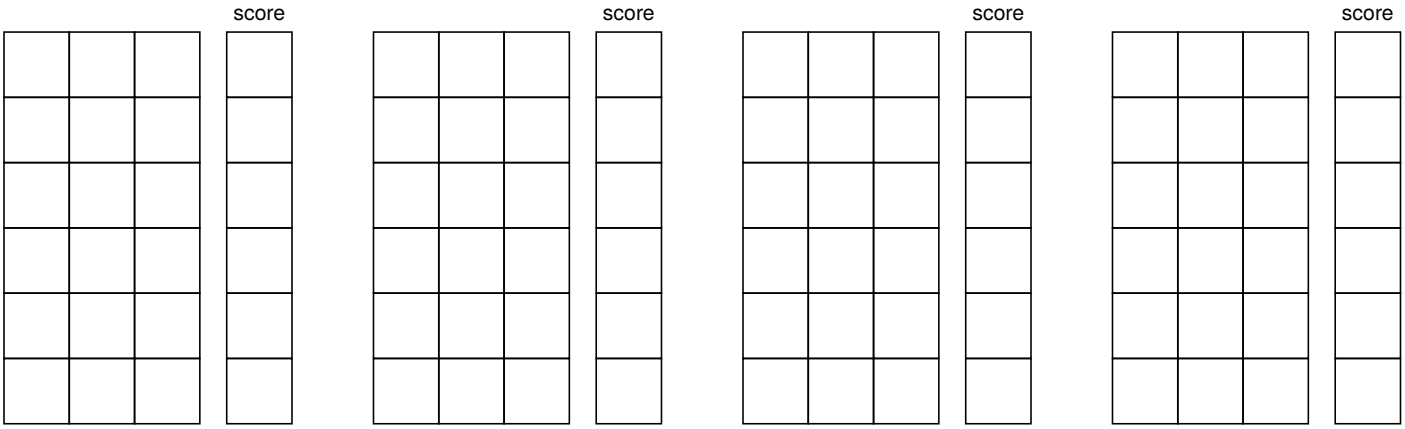
			score
R	R	R	2
R	Y	Y	2

				score
R	R	Y	Y	3
Y	R	Y	R	1
R	R	R	Y	2

				score
R	R	Y	Y	2
R	Y	R	Y	2
R	Y	Y	R	2

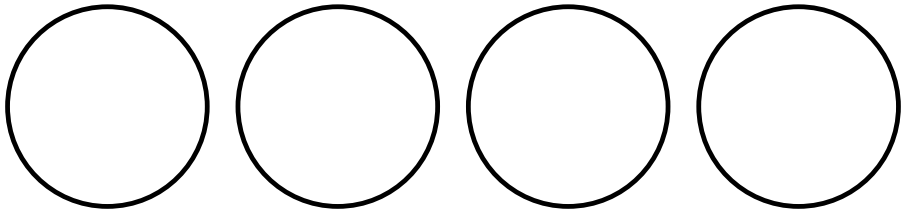
				score
R	Y	R	Y	1
R	R	Y	Y	3
Y	Y	Y	Y	2

Problem 4: Find a way to choose your tests so that you always know a three-counter secret after three tests or fewer. Describe it so that someone else could use it.

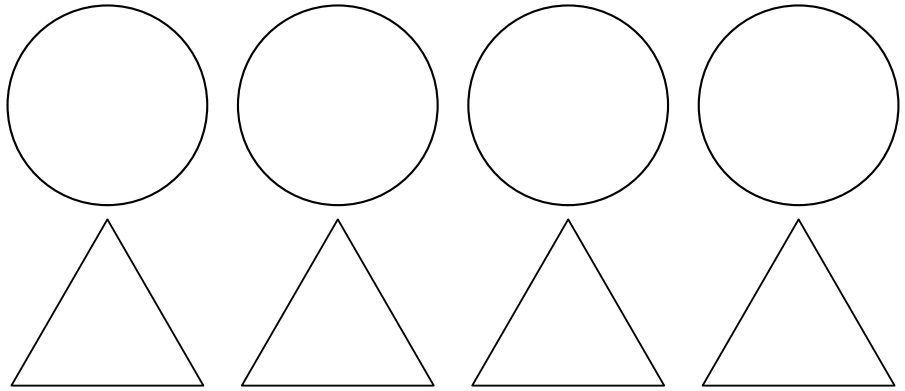


Problem 5: Play the code game with four counters, taking turns. Find each secret with as few tests as you can.

secret



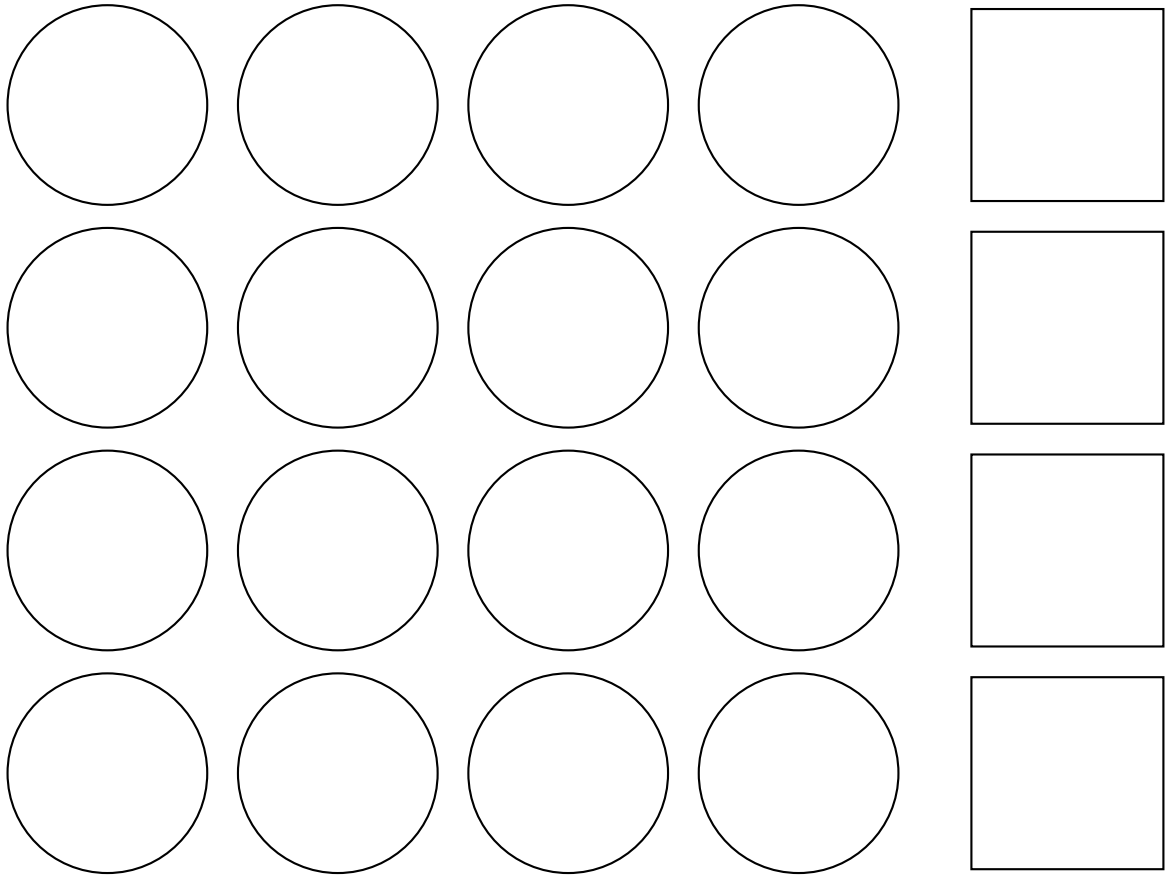
copy of test



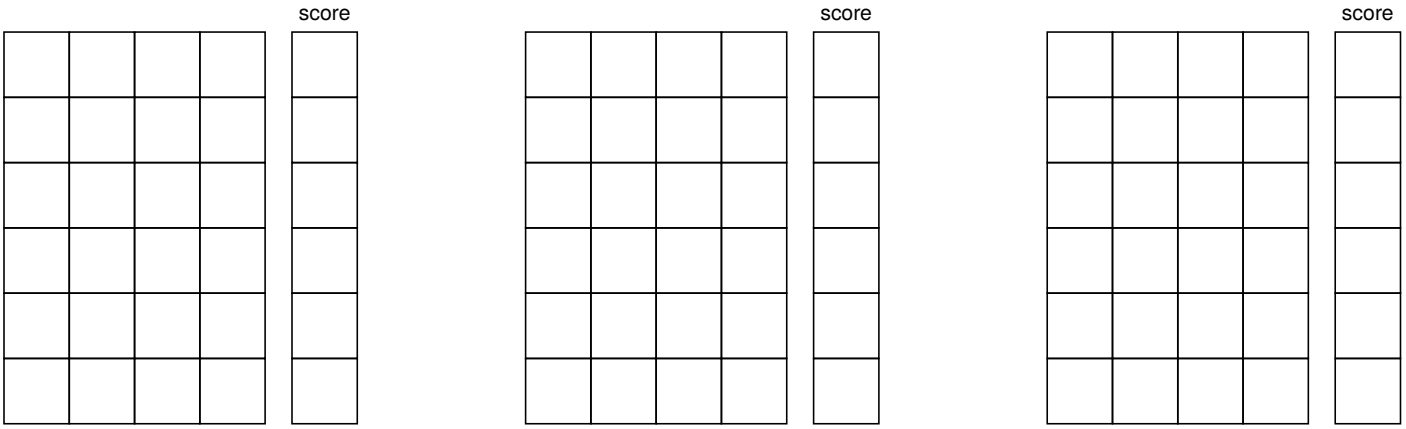
folder

score

test

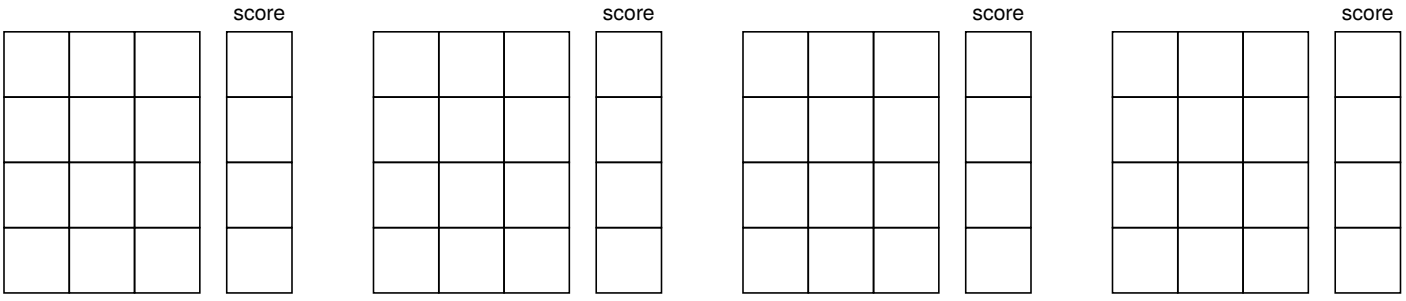


Problem 6: Find a way to choose your tests so that you always know a four-counter secret after four tests or fewer. Describe it so that someone else could use it.

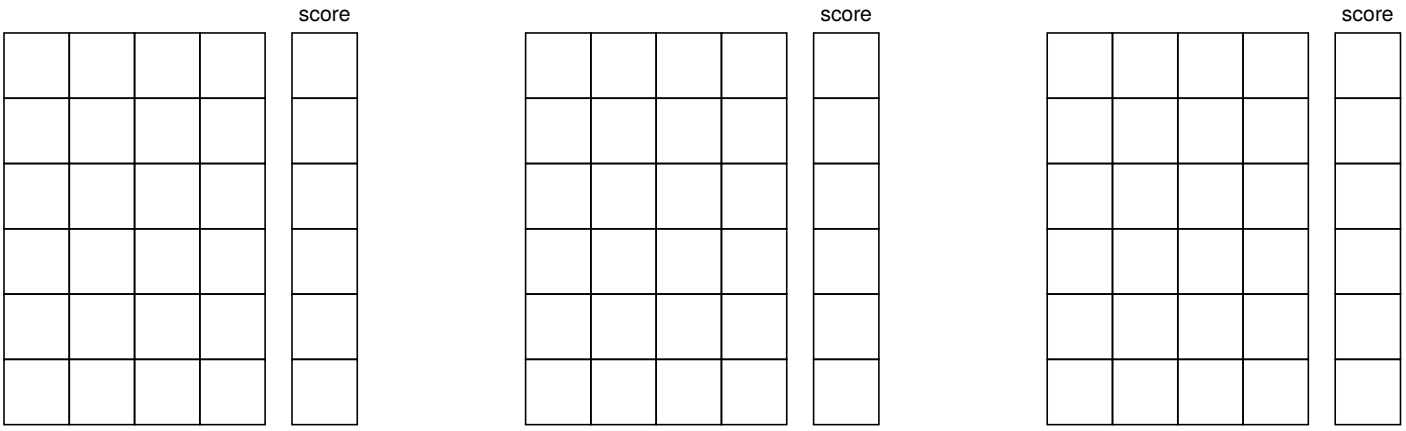
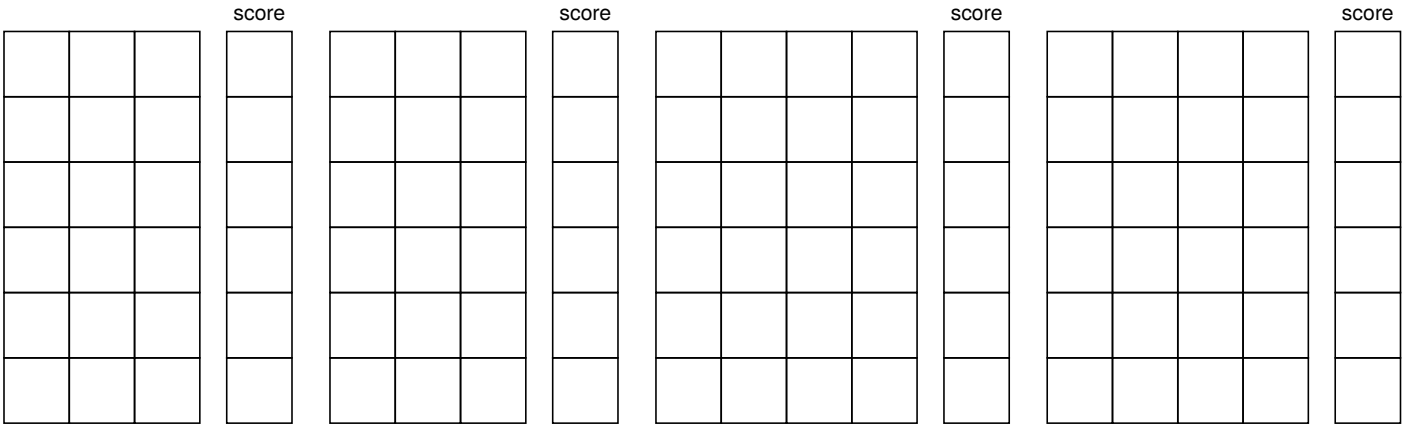


Problem 7: Lay out any secret row and any test row, one under the other, and find the score. Then turn over the two counters in the first place, one in each row. What happens to the score? Will that happen whatever the two rows are and whichever place you turn over? Explain.

Problem 8: Can one test always be enough to be sure of a two-counter secret? Can two tests always be enough to be sure of a three-counter secret? Explain.



Problem 9: Can three tests always be enough to be sure of a four-counter secret? Explain.



Problem 10: Change the three-counter game so that it ends only when a test scores 3. How many tests does the breaker need to be sure of ending the game, whatever the secret is? Answer the same question for four counters, where the game ends when a test scores 4. Explain both answers.